Multimodal Discriminative Feature Learning for SAR ATR: A Fusion Framework of Phase History, Scattering Topology, and Image

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***Abstract*— Deep learning has emerged as the dominant paradigm for synthetic aperture radar (SAR) automatic target recognition (ATR), which induces discriminative visual features from target images. Due to the imaging mechanism, the SAR image is a reconstruction of the interaction between the electromagnetic wave and the target, which intrinsically entangles the radar characteristics, target scattering properties, and visual signatures. However, current learning algorithms focus primarily on the visual signatures without explicit specification or ineffective integration of the full range of SAR features into the learning process, leading to a weak generalization ability across different radar systems and imaging conditions. To address this issue, we propose a novel multimodal feature fusion learning framework, which captures a comprehensive set of target features from different domains for enhanced complementarity. First, we encode the target response and radar characteristics into the phase-history data. A cross-direction sequence learning module is designed to extract their range and azimuth dependence. Next, a hierarchical graph node-aggregating neural network is developed to learn the scattering topology features from scattering points according to the part-to-whole learning bias. Finally, the learned features from the phase-history and scattering domains are fused with the image features obtained from an off-the-shelf deep feature extractor for final target recognition. Experiments on the moving and stationary target acquisition and recognition (MSTAR) benchmark demonstrate its effectiveness. Compared with the other SAR ATR algorithms, our approach can achieve the state-of-the-art recognition accuracy without using any data augmentation trick, especially in cases of limited training samples.**

***Index Terms*— Automatic target recognition (ATR), feature fusion, phase-history data, scattering topology, synthetic aperture radar (SAR) characteristics.**

I. Introduction

SYNTHETIC aperture radar (SAR) is a remote sensing technology that has found diverse applications in both civilian and military domains due to its ability to operate in adverse weather conditions, provide uninterrupted coverage over long ranges, and deliver high-resolution imagery [1]. As a result, automatic target recognition (ATR) using SAR data has emerged as a critical research area to translate the data into actionable intelligence. However, the unique electromagnetic imaging mechanism of SAR, which relies on the interaction between the radar signal and the target, poses significant challenges for ATR. This is because SAR data are contingent on the physical characteristics of the target, the radar parameters, and the imaging geometry. It necessitates the development of specialized ATR algorithms that can accurately model the intricate scattering behavior of targets from SAR data.

SAR ATR system typically involves two principal stages: feature extraction and classifier construction [2], [3]. The goal of feature extraction is to transform the raw SAR data into a set of meaningful features that can be used to distinguish between different targets, while the goal of classifier construction is to develop a decision rule that maps the extracted features to the corresponding target labels. Therefore, the representability and discriminability of the features strongly influence the recognition performance. Currently, there are two main approaches to SAR target extraction. The first approach studies SAR target data from physical electromagnetic scattering mechanisms. A parametric attribute scattering center (ASC) model derived from the geometric theory of diffraction (GTD) theory [4], [5], [6], [7], [8] suggests that high-frequency radar measurements of man-made targets are dominated by returns from multiple isolated scattering centers, each of which can be characterized by some parameters reflecting the relevant physical scattering mechanisms. In this sense, the target is composed of some combinations of these scattering centers, and the set of parameters can serve as target features. By collecting and computing the parameter sets of targets from different acquisition conditions, such as different aspect angles and depression angles, a target-specific feature template can be constructed. The query test target is then recognized by pairwise template matching. However, these deductive model-based features are affected by practical disturbances such as speckle noise and the surrounding environment. Although the ASC model can accurately describe an SAR target in a physically plausible way, the exact computation of the parameters is always an intractable problem in practical situations because it requires solving an ill-posed inverse problem that is sensitive to noise and model errors.

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In addition to scattering features, data-based approaches involve manually extracting data features, such as statistical signal signatures and visual image features, in a feed-forward manner [9], [10], [11], [12], [13], [14], [15]. These features are more robust than scatter features because they can capture more information-rich characteristics of the target. However, the features are often extracted without considering their identity information, resulting in less discriminative features and less effective classifiers. To address this issue, recent ATR research has shifted to an end-to-end deep learning-based framework that integrates feature extraction and classifier construction into a unified learning paradigm. This approach enables the simultaneous learning of features and classifiers, resulting in significant improvements in ATR performance [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. Moreover, in order to test the system in the open world, Openmax and modified Openmax classifiers have recently been introduced in SAR and inverse SAR ATR tasks to enable convolutional neural networks (CNNs) to differentiate between open and closed set classes [28], [29], [30]. Nevertheless, this inductive learning-based recognition paradigm suffers from poor generalization ability [31], especially when SAR target training data are insufficient. Furthermore, the learned features are often poorly interpretable and challenging to provide credible and actionable decisions in practical applications.

To overcome the above challenges, recent research has focused on fusing model-based and data-based features. This fusion approach aims to take advantage of both deductive and inductive learning features to improve recognition performance. Ding et al. [32] proposed a decision-level fusion framework for SAR ATR, which simultaneously considers the ASC and image features during recognition. Zhang et al. [33] focused on concentrating a bag of ASC features with deep image features, with which a discriminative classifier is learned. However, it is not an end-to-end framework due to bag-of-words operations, which may limit the discriminative power. More recently, Liu et al. [34] developed an end-to-end framework for multilevel scattering center and deep feature fusion learning. This approach considers ASC features as set data and uses deep set-based algorithms for further learning. However, the performance of this method is strongly influenced by the accuracy of the computed ASC parameters.

A. Motivation

Despite the promising results achieved by recent works that fuse physical ASC model-based and data-based features for SAR ATR, a significant amount of SAR information remains unutilized in these approaches, thereby limiting their generalization performance. Considering the unique electromagnetic imaging mechanism of SAR, the target image is a reconstruction of the data samples in the phase-history domain [35], [36], which reflects the interactions between the target and the transmitted radar signal. Therefore, SAR data entangle both the radar characteristics and the target signatures, making it difficult for ATR methods to generalize across different radar systems with varying characteristics such as bandwidth, center frequency, polarization, and imaging geometry. This is because these methods rely heavily on the fundamental independent identity distribution assumption, which hinders their ability to generalize well to new target data and limits their effectiveness in real-world applications. We assert that SAR target data contain additional radar-related features beyond the currently used ASC and image features, which, if effectively exploited, can improve the accuracy, reliability, and generalizability of recognition performance. Failure to leverage these radar-specific characteristics in the learning framework may limit its ability to understand SAR target data and cause its inductive learning direction to deviate from that of human understanding. Currently, few learning-based SAR ATR algorithms consider the radar characteristics during discriminative learning.

In addition, previous SAR ATR algorithms use ASC features to capture the local signatures of the scattering centers, accounting for the local subcomponents of the target, but fail to capture the global topology among different scattering centers without considering their interactions. The global topology among the scattering centers is a critical feature that reflects the spatial arrangement of the subcomponents of the target and the interaction between them. Effective modeling of the global topology can provide complementary information to the ASC features, leading to improved performance in SAR ATR. Therefore, the inability of previous algorithms to capture the global topology among different scattering centers highlights the need for a new approach that can effectively exploit this feature.

In summary, the above two perspectives directly motivates us to develop an SAR ATR model that can effectively exploit the available radar-related features and the global scattering topology in SAR target data. Such a model has the potential to provide more accurate, reliable, and generalizable recognition performance for a single or across different radar systems.

B. Contribution

This article presents a novel multimodal feature fusion learning framework, named MoFFL, for SAR ATR. The proposed framework aims to capture a comprehensive set of radar characteristics, scattering topology, and image features from different target data domains and achieve feature fusion for enhanced complementarity. To achieve this goal, this article introduces an encoding scheme that transforms each phase-history data along the range direction into a raw feature vector sequence containing both the target response and radar characteristics. Based on the vector sequences at different observation angles, a cross-direction sequence learning module is designed to transform all vector sequences into range direction feature embeddings and capture their azimuth dependence as discriminative features of the phase-history domain. Furthermore, to learn the discriminative scattering topology feature from a set of irregular scattering points, a hierarchical graph node aggregating neural network is developed that models and clusters the scattering topology of the points layer by layer in a part-to-whole manner. Finally, the phase-history domain and scattering topology features are fused with the image features obtained from an off-the-shelf deep feature extractor backbone for final target recognition. The proposed

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approach is evaluated on the moving and stationary target acquisition and recognition (MSTAR) benchmark. The experimental results demonstrate the effectiveness of the proposed approach. Compared with other ATR algorithms, to the best of authors' knowledge, the proposed framework achieves the state-of-the-art recognition accuracy in the case of limited training samples without using any data augmentation trick. The contributions of this work are summarized as follows.

1) We propose a novel end-to-end multimodal feature learning framework that simultaneously learn and fuse the discriminative features from the phase history, scattering points, and SAR image. We show that these three different modalities, respectively, contains SAR characteristics, scattering topology, and visual features of the target so that it achieves enhanced complementarity and performance in comparison with the previous ATR approaches with a single image or ASC modality.
2) We declare that SAR target data entangle the additional radar-related features beyond the currently used ASC and image features, which should be exploited to improve the accuracy, reliability, and generalizability. We developed an effective cross-directional sequence learning module to learn features related to the target response and radar characteristics including bandwidth, centering frequency, and observation angle. This module treats the radar characteristic-encoded history data as bidirectional vector sequences along the range and azimuth directions, and has demonstrated promising results in learning these features.
3) We construct a hierarchical graph node aggregating neural network for scattering topology features from the set of irregular scattering points. This module models and clusters the scattering topology of points layer by layer, thereby enabling the learning bias of subcomponents to parts to the whole.

The rest of this article is organized as follows. Section II introduces the preliminary and related works. Section III formally proposes the main framework and its detailed architecture. Extensive experiments are carried out in Section IV to demonstrate the effectiveness and superiority of the proposed model. Section V summarizes this work and suggests future directions.

II. Preliminary and Related Works

In this section, we will provide a preliminary overview of these two research areas and discuss relevant works in the literature to further highlight our insightful motivation. Specifically, we will first introduce the phase-history data in spotlight mode SAR and its potential utility for ATR. We will then discuss the physical-based ASC model and their effectiveness in SAR ATR task.

A. Phase-History Data of Spotlight Mode SAR

Spotlight mode SAR has become a prominent technology for ATR, utilizing microwave radiation to produce high-resolution images of the target [35], [36]. As depicted in Fig. 1, the SAR platform travels along a flight path, while emitting periodic pulses of microwave radiation, typically in the form of a linear frequency-modulated (LFM) signal (the following equation), at observation angle θ_i and depression angle φ toward targets within an illuminated area S:

$$s(t) = \exp\{j[\omega_0 t + \alpha t^2]\},\ t \in \left[-\frac{T_p}{2}, \frac{T_p}{2}\right] \tag{1}$$

where ω_0 denotes the center frequency, 2α corresponds to the chirp rate, and T_p denotes the pulsewidth. The emitted pulses are subsequently reflected back to the radar and demodulated to generate the phase-history data, given by the following equation:

$$r(t,\theta_i) = \int_{-L}^{L} p_{\theta_i}(\bar{n}) \exp[-j\bar{n}\Omega(t)]\mathrm{d}\bar{n} \times t \in \left[-\frac{T_p}{2} + \frac{2(R_0+L)}{c}, \frac{T_p}{2} + \frac{2(R_0-L)}{c}\right] \tag{2}$$

where $p_{\theta_i}(\bar{n})$ defined in the following equation is the range profile representing the integrated reflectivity function of $X(m,n)$ from observation angle θ_i:

$$p_{\theta_i}(\bar{n}) = \int_{-L}^{L} X(\bar{m}\cos\theta_i - \bar{n}\sin\theta_i, \bar{m}\sin\theta_i + \bar{n}\cos\theta_i)\mathrm{d}\bar{m} \tag{3}$$

$\Omega(t) = (2/c)(\omega_0 + 2\alpha(t - (2R_0/c)))$ is the spatial-frequency variable, $X(m,n)$ represents the microwave reflectivity density function of the illuminated target scene, namely, the unknown complex SAR image to be reconstructed and the c is the speed of light. In this sense, the range distance is "converted" to the spatial frequency. During the flight, the spotlight mode SAR remains focused on the target region and captures a sequence of phase histories $\{r(t,\theta_i)\}_{i=1}^{K}$ from K continuous viewing angles $\{\theta_c \pm k\Delta\theta\},\ k = 1,\ldots,K/2$ with interval $\Delta\theta$. Consequently, the collection of phase histories traces out a ribbon surface whose precise shape is determined by the flight trajectory shown in Fig. 2. It is commonly assumed that the depression angle φ remains constant within the straight-line trajectory. Under this assumption, the ribbon surface is a plane, and it exists in the spatial-frequency domain, commonly referred to as the phase-history domain [36]. The discrete phase-history data can be assembled as a polar collection raster $\{\mathbf{r}(\theta_1),\ldots,\mathbf{r}(\theta_K)\}$ by sampling each $r(t,\theta_i)$ according to the Nyquist sampling law to obtain a complex vector $\mathbf{r}(\theta_i) \in \mathbb{C}^{N_r}$.

In this domain, each sample point distributed on a circular annulus is associated with a specific pair of frequency and angle as the coordinates. Each radial segment accounting for the range of spatial frequency has the same length for each emitted pulse, and its value range is approximately determined by the time support as

$$\Omega \in \left[\frac{2}{c}(\omega_0 - \pi B_c), \frac{2}{c}(\omega_0 + \pi B_c)\right] \tag{4}$$

where B_c is the bandwidth of the chirp in radians. Accordingly, the entire annulus is offset from the origin by a specific amount of $2\omega_0/c$ so that we can conveniently compute the approximate frequency values for the points in each radial segment. This

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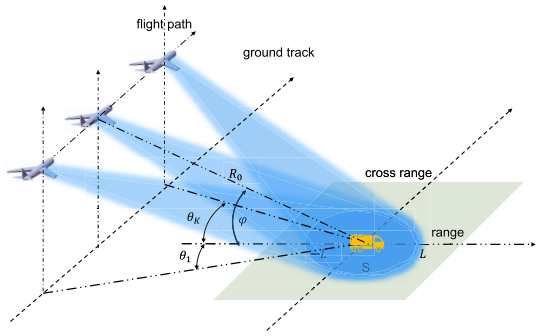


Fig. 1. Illustration of the SAR imaging geometry.

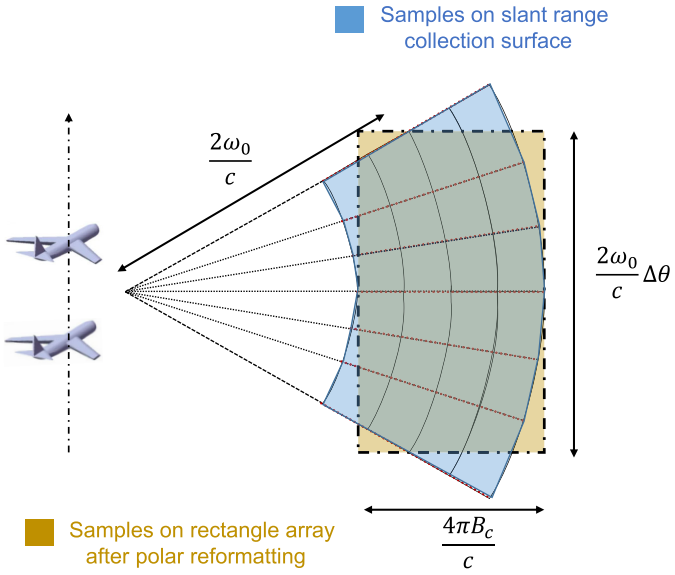


Fig. 2. Illustration of the samples in the phase-history domain.

allows for the interpretation of the phase-history samples as the responses of the target at specific frequency and observation angle, which can directly contribute to the performance of SAR ATR. Despite the potential utility of this information, few existing SAR ATR algorithms have been developed to capture the underlying target-specific patterns present in the phase-history domain, with most works focusing on recognition with the 1-D range profile [9], [13], [37], [38], [39] or 2-D image [14], [15], [16], [21], [23], [24], [25], [26], [27], [33], [40], [41]. To facilitate image reconstruction from phase-history data, polar-to-rectangular interpolation, shown in Fig. 2, is a widely used technique to generate regularly spaced samples on a Cartesian grid from the irregularly sampled phase-history data. The interpolated samples are then transformed into an image using inverse discrete Fourier transform. It is worth noting that the spotlight SAR phase history can be considered a hologram since any part of it can be used to reconstruct an image of the entire scene using a subaperture imaging strategy. However, reconstructed images of the same target may correspond to different responses in frequency and observation angle in the phase-history domain, despite their visual similarity shown in Fig. 3. To investigate this issue, we conducted an ATR experiment using the baseline model A-ConvNet [16]. We used target images reconstructed from

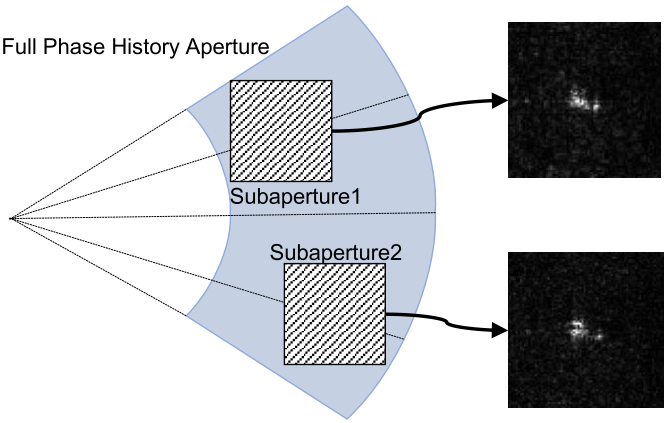


Fig. 3. Holographic property of spotlight SAR imagery.

different parts of the phase-history data as training and test samples. Ideally, the recognition accuracy should be close to 100% since the samples are obtained from the same phase-history data. However, the recognition accuracy for both 15° and 17° depression angles was only about 95%, indicating that the features extracted from SAR images alone are not sufficient to reflect the target characteristics. Therefore, phase-history data contain more comprehensive information than visual signatures in the image domain. However, current image-based ATR algorithms primarily disregard this information, resulting in a limited ability to capture target-specific patterns present in the phase-history domain. This restriction impedes the ability to train these recognition algorithms with SAR images from one frequency band and apply them to images from another frequency band. As a result, these algorithms cannot be used interchangeably across radars of different bands, limiting their potential practicality in diverse operational scenarios.

Worrell and Soumekh [42] proposed a signal subspace matched filtering technique to exploit phase-history data for SAR ATR, but no empirical evaluation of the proposed method was conducted on simulated or real SAR data, leaving the effectiveness of the approach uncertain. In [43], off-the-shelf CNNs, namely, VGG16 [44] and AlexNet [45], were directly applied to classify the 2-D phase-history data. However, the results did not demonstrate improved performance compared to those based on target images with the same deep networks. Consequently, the effective utilization of phase-history data for discriminative feature extraction/learning poses a significant challenge, resulting in the underlying target-specific patterns in the phase-history domain being largely unexplored. They did not utilize the radar signature in the phase-history domain as well. To address this challenge, this study proposes a novel model for SAR ATR that explores the potential of phase-history data. The proposed model demonstrates a remarkable superiority, highlighting the significance of exploiting phase-history data for improving the recognition performance.

B. Attribute Scattering Centering-Based ATR

In addition to image-based SAR ATR methods, GTD-based ASC model also plays a critical role [4], [5]. While

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image-based feature learning methods have gained dominance in the SAR ATR domain, scattering model-based features offer greater physical interpretability and enable a human credible decisions during recognition. It suggests that high-frequency radar measurements of man-made targets are dominated by returns from several scattering centers, given by

$$r(\omega,\theta) \approx \sum_{c=1}^{C} A_c \left(\frac{j\omega}{\omega_c}\right)^{\alpha_c} \exp\left(\frac{-2j\omega}{c}(x_c\cos\theta + y_c\sin\theta)\right) \cdot \operatorname{sinc}\left(\frac{\omega}{c}L_c\sin\left(\theta-\bar{\theta}_c\right)\right)\exp(-\omega\gamma_c\sin\theta) \quad (5)$$

where the model parameters $\{A_c, \alpha_c, x_c, y_c, L_c, \bar{\theta}_c, \gamma_c\}$ characterize the scattering property depending on the frequency ω and observation angle θ. For cth scatter center, A_c is the amplitude, α_c is the frequency dependence, L_c and $\bar{\theta}_c$ are the length and orientation, respectively, x_c, y_c are its location, and γ_c is the aspect dependence on ASC. By comparing (5) with (2), it is evident that the ASC model can be considered as a physical representation model of the phase-history data, which assists in the extraction of local target scattering properties. This observation emphasizes the fact that the phase-history data comprises multiple centers that reflect canonical scattering mechanisms, such as dihedral and trihedral, which are responses of local subcomponents of the target. Therefore, the scattering centers contain valuable physical and geometric information about these subcomponents, enabling an accurate description of the target scattering behavior in a part-to-whole manner.

In general, scattering centers can be classified into two categories: localized or distributed. Localized scattering centers, such as trihedral and top-hat reflection, are aspect-invariant and have model parameters $L = 0$ and $\bar{\theta} = 0$. Distributed scattering centers, such as dihedral and cylinder reflection, correspond to lines or symmetrical dots in SAR images and have model parameters $\gamma = 0$, $L > 0$, and $\bar{\theta}$ measuring their orientation. Therefore, it follows that the model parameters reflect the scattering type and, therefore, are usually utilized as the scattering feature of the target for template-based recognition [4], [5] or fused with deep image features [33], [34]. However, it is difficult to determine the number of scattering centers for an unknown target, yielding the parameters-estimation task an ill-posed inverse problem. Greedy methods or genetic algorithms are commonly employed as a means of searching for a compromise solution, with which a template matching classifier is exploited for subsequent recognition. Chiang et al. [6] proposed a Bayesian-based scattering center feature matching method. Dungan and Potter [7] evaluated the least trimmed square Hausdorff distance for scattering center set similarity using a particle swarm approach. Zhou et al. [5] matched ASCs for different hierarchical regions for decision. Fu et al. [8] proposed a template matching-based aircraft identification method with scattering structure features.

In recent years, there has been a shift in focus from the exact computation of model parameters to the characterization of scattering topology [46]. This approach is based on the premise that the scattering topological properties of a target can be roughly characterized by the distribution of its scattering centers, or topology. By exactly analyzing the topology of strong scattering points, it is possible to extract invariant or equivariant topological features that are independent of the target's pose or pitch angle. Compared to traditional approaches that rely on accurately computing the ASC parameters for each center, the strong scattering points can be easily determined from the target image by sorting, providing several advantages. However, it is important to note that this approach does not capture the hierarchical scattering topology that represents the features of subparts to the whole. To address this limitation, this study will propose a hierarchical scattering topological feature learning model that is capable of gradually capturing the local to global scattering features of the target. By leveraging the hierarchical nature of the scattering topology, our proposed model can effectively extract the subcomponent-level features and integrate them to form the global features of the target. This enables our model to achieve more comprehensive and accurate target recognition results.

III. Multimodal Discriminative Feature Learning Framework

This section formally develops the MoFFL framework for SAR ATR, which aims to learn discriminative features from multiple modalities, including the phase-history, image, and scattering topology domains, and fuse them for target recognition. To address the challenge of handling multimodal data in the form of polar format, isolated scattering point set, and 2-D image, we propose a cross-direction sequence learning module for feature extraction from the phase-history domain, which captures the correlation between data points in the range and cross-range directions. In addition, we design a hierarchical graph node-aggregating neural network to extract scattering topological features from strong scattering points of the target. Furthermore, we extract image features using a baseline deep learning network as the backbone. Finally, we integrate these features end-to-end as complementary supervisory information for the final SAR ATR. The overall architecture of MoFFL is illustrated in Fig. 4.

A. Phase-History Feature Extraction Module

As outlined in Section II-A, phase-history data are typically composed of K range radial segments, each corresponding to a particular observation angle. Each sample point in a range radial segment reflects the change in response of the incident electromagnetic wave frequency, while the change of time in the cross-range direction reflects the change of radar observation angle. In order to extract discriminative features from the phase-history domain, a cross-direction sequence learning module is proposed. This module employs a divide-and-conquer and integration scheme for feature extraction. Specifically, it first extracts the feature of each range radial segment in parallel along the range direction. Subsequently, the range features of these K segments are integrated to extract the output feature by considering their dependence along the azimuth direction.

We begin by examining the initial representation of each sample point in each range radial segment. In [43], the

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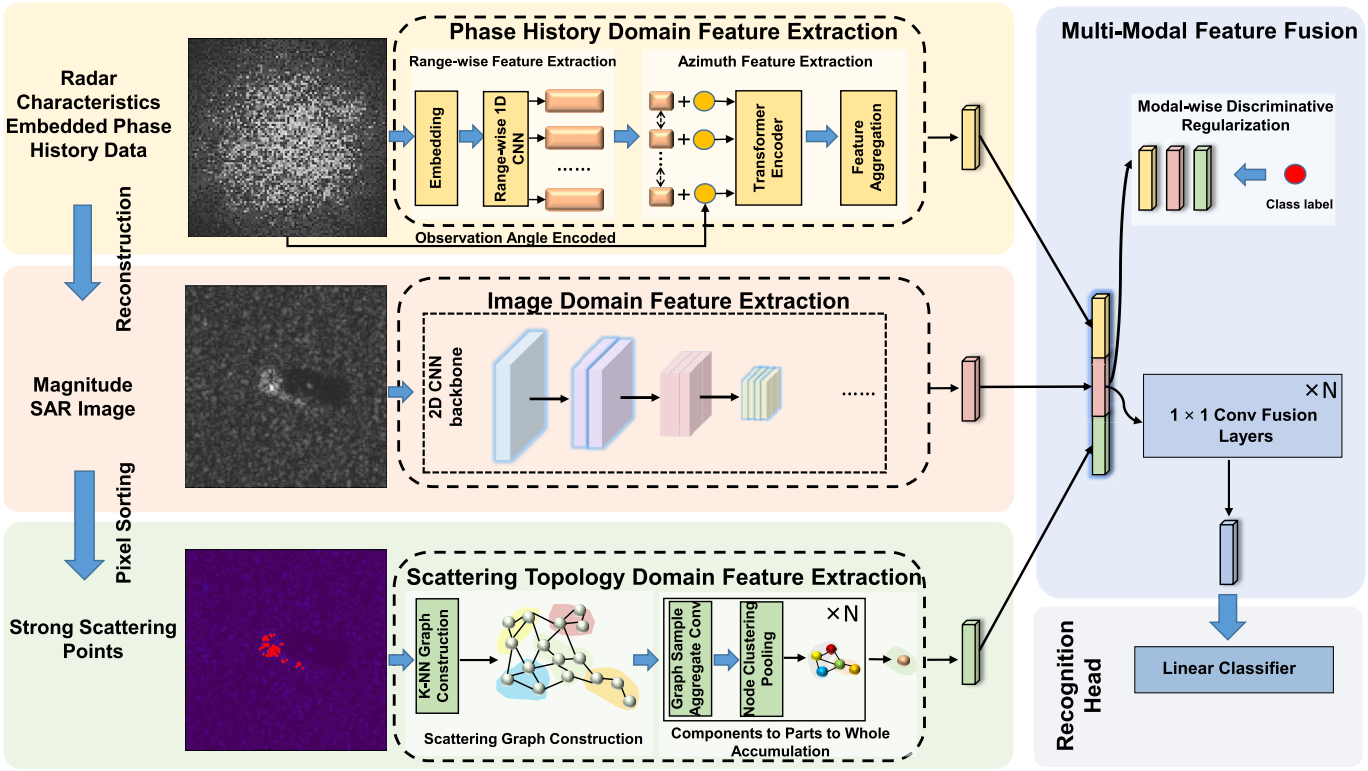


Fig. 4. Overall pipeline of the multimodal discriminative feature learning framework for SAR ATR.

phase-history data were transformed into a 2-D array and processed using a standard 2-D CNN model designed for image classification. However, this approach has two limitations. First, the 2-D phase-history array only utilizes the magnitude response of the data, while discarding the phase information that is a crucial component of the electromagnetic properties of the target. Moreover, the coordinates of the image data array only reflect the positions of the pixels without any additional physical meaning. Consequently, a general 2-D CNN model mainly focuses on extracting the correlations among pixels in a local neighborhood. In contrast, the coordinates in the phase-history data array correspond to the spatial frequency and observation angle, which are crucial for mining the target's electromagnetic properties. Ignoring the dependence of each data point on spatial frequency and observation angle can lead to a loss of critical electromagnetic characteristics of SAR. Therefore, it is essential to account for these factors when processing the phase-history data. Second, the 2-D convolution operation employed by the CNN model can only capture local neighborhood correlations of data points in the range and azimuth directions, while the phase-history data exhibit long-term correlation, particularly in the azimuth direction. Each range radian segment corresponds to the target's response at different observation angles, so the data across different radian segments should exhibit pairwise correlation properties. These properties cannot be captured using 2-D CNNs. Thus, a different approach to processing the phase-history data is required that can capture the long-term correlation of data points in the azimuthal direction and incorporate the spatial frequency and observation angle correlation properties of each radian segment.

Based on the above analysis, a sample point encoding the radar signatures can be represented as a triplet (r, ω, θ), where r is its complex value and corresponds to the response at a specific frequency ω and angle θ of the target scattering center. To enable further analysis, this triplet is represented as a 4-D vector $\mathbf{r} = [10\log(|r|+1), \sin(\angle r), 2\omega/c, \sin(\theta)]^{\mathrm{T}}$, where $|\cdot|$ and $\angle$ denote the magnitude and phase angle of r, respectively. The corresponding ω and θ of each point can be calculated based on the radar parameters. To simulate and compensate the error during polar-to-rectangle interpolation and the computation, we randomly sample ω and θ from $\mathcal{N}(\omega, \sigma^2)$ and $\mathcal{N}(\theta, \sigma^2)$, respectively, where σ^2 is a hyperparameter accounting for the error variance. Using this representation, each range radial segment $\mathbf{r}(\theta_i)$ can be viewed as a data sequence containing N_r random vectors, and range feature extraction becomes a sequence-to-vector task. To address this task, 1-D CNN, recurrent neural networks (RNNs), and long short-term memory (LSTM) [25], [37], [39] are commonly used methods. In this article, we prioritize efficient parallelization of computation and use 1-D CNN to extract range features from the phase-history data. We first embed the 4-D vectors of each phase-history data using a linear transformation. Then, for each embedded vector along the range direction, we obtain the filtered feature responses captured at different observation angles by sharing the 1-D convolution operation. The features will be aggregated for each feature response using the max-pooling layer. Finally, by stacking these 1-D convolutional and maxpooling operations layer by layer, the input sequential random vectors of every radial segment are encoded into a d-dimensional feature vector $\{\mathbf{f}_{\theta_i} \in \mathbb{R}^d\}_{i=1}^{K}$, respectively.

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Our next aim is to capture the pairwise dependence of K range feature vectors along the cross-range direction. To achieve this, we use the transformer encoder [47], which uses a multihead attention mechanism to measure the pairwise correlations among the K range feature vectors. More formally, we can arrange the set of $\{\mathbf{f}_{\theta_i}\}_{i=1}^{K}$ into a matrix $\mathbf{F} = [\mathbf{f}_{\theta_1}, \ldots, \mathbf{f}_{\theta_K}]^{\mathrm{T}} \in \mathbb{R}^{K\times 128}$. Instead of using a single attention function, the transformer encoder employs a multihead attention operator (6) to attend to the information of observation angles from different subspaces

$$\begin{aligned}\mathrm{MultiHead}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \mathrm{Concat}(\mathrm{head}_1,\ldots,\mathrm{head}_h)\mathbf{W}^O \\ \text{s.t. } \mathrm{head}_i &= \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{W}_i^Q\cdot\left(\mathbf{K}\mathbf{W}_i^K\right)^{\mathrm{T}}}{\sqrt{d_i}}\right)\mathbf{V}\mathbf{W}_i^V\end{aligned} \tag{6}$$

where $\mathbf{W}^O \in \mathbb{R}^{d\times d}$, $\mathbf{W}_i^Q \in \mathbb{R}^{d\times d_i}$, and $\mathbf{W}_i^K \in \mathbb{R}^{d\times d_i}$ and $\mathbf{W}_i^V \in \mathbb{R}^{d\times d_i}$ are three linear parametric projection matrices in ith head, and $\sum_i d_i = d$. The transformer encoder utilizes a multihead attention mechanism to enable us to attend to different subspaces of the input range feature vectors, thereby allowing us to measure the pairwise correlations among the K vectors. To perform this operation, we use a self-attention mechanism in which the query, key, and value matrices are set to be equal to the input matrix, $\mathbf{Q},\mathbf{K},\mathbf{V} = \mathbf{F}$. The attention operator projects each range feature vector onto two different subspaces using $\mathbf{W}^Q$ and $\mathbf{W}^K$ and then weights their subspace representations, $\mathbf{F}\mathbf{W}^V$, using the normalized inner product between them. This inner product-based attention mechanism is more effective at capturing long-range correlations than RNN and LSTM models, as it can capture both local and global correlations and avoid the problem of weak long-range correlation capture typically encountered in these models. Following subsequent layer normalization and anglewise transformation with a feed-forward network, the transformer encoder outputs a representation for each range radial segment, namely, $\mathbf{z}_{\theta_i} \in \mathbb{R}^d$. To incorporate observation angle information into the input of each transformer encoder layer, we add the sine of the observation angles, $[\sin(\theta_1),\ldots,\sin(\theta_K)]^{\mathrm{T}}$, to $\mathbf{F}$. Unlike the typical transformer model, which employs sine and cosine functions to encode even and odd positions, respectively, this modification is necessary because the spotlight mode SAR platform operates within a relatively small range of observation angles, and encoding the observation angle information explicitly in the input range feature vectors improves the transformer encoder's performance. The global average pooling used as the feature aggregate operator is performed on $\mathbf{Z} = [\mathbf{z}_{\theta_1},\ldots,\mathbf{z}_{\theta_K}]^{\mathrm{T}} \in \mathbb{R}^{K\times d}$ for information integration from all observation angles, yielding the final phase-history feature vector $\mathbf{f}_p \in \mathbb{R}^d$

$$\mathbf{f}_p = \mathrm{AGGREGATE}\left(\mathbf{z}_{\theta_1},\ldots,\mathbf{z}_{\theta_K}\right). \tag{7}$$

B. Scattering Topology Feature Extraction Module

Scattering center-based models are crucial for SAR ATR, as they reflect the physical plausible structural information of the target [4], [5]. However, accurately computing the entire model parameters can be challenging. Alternatively, we simplify the ASC model by only considering the partial parameters, namely, A_c, x_c, y_c. More formally, the peak of the scattering region, or strong scattering points, on the SAR magnitude image provides a simpler and more efficient approach to identify the dominant subcomponents of a target. These points mostly account for the dominant subcomponents of a target, so their spatial distributions, contained in $\{x_c, y_c\}$, can reflect the geometric topology between the target subcomponents. Therefore, the underlying regularity of the consistent topology of a target carries target-specific characteristics that can be used for recognition.

To extract discriminative scattering topology features, we first identify the strong scattering points by sorting the magnitude of the pixels in an SAR target image and selecting the top N pixels. However, learning from these unstructured and unregularized points is challenging compared to learning from structured phase-history data and images. To address this challenge, constructing an undirected graph $\mathcal{G}^0 = \{V, E\}$ with these points and applying graph learning algorithms for scattering topological feature extraction is a promising approach [48], [49]. In this graph, the vertex set $V = \{v_1,\ldots,v_N\}$ represents the strong scattering points as nodes, and their connected relations are constructed as edges in E. To construct the graph, the node features should be specified first. Following the simplified ASC model, the position parameters $\{x_c, y_c\}$ for each scattering point can be conveniently computed, and A_c can be approximated by the corresponding magnitude of the pixels. Thus, the initial node features are designed as $\mathbf{v}_c = [A_c, x_c, y_c]^{\mathrm{T}}$, which are arranged into a node feature matrix $\mathbf{V} \in \mathbb{R}^{N\times 3}$. To determine the edges between the nodes, the K-nearest neighbor (KNN) algorithm is utilized to select several nodes as neighbors for each node based on their Euclidean distance, where two nodes are connected if their corresponding strong scattering points are within a certain spatial distance threshold. According to this principle, the edge set E is built, along with an adjacency matrix $\mathbf{A}$ given by

$$\mathbf{A}_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in E \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

The resulting adjacency matrix is used to define the connectivity structure of the graph. By constructing the graph and modeling the strong scattering points as nodes, an appropriate graph learning method will be utilized to extract discriminative and structured features from the scattering topology domain.

To extract discriminative and structured features from the constructed graph $\mathcal{G}$, several popular graph neural network (GNN) frameworks, such as the graph convolution network [50] and the graph attention network [51], could be used as a feature extractor backbone. However, these frameworks are not suitable due to the following limitations. First, they are inherently flat without capturing the hierarchical structure of the graph. In our task, we construct a graph composed of strong scatter points corresponding to the subcomponents of the target. We have a bias toward inductive learning, aiming to learn the geometric topological relations of the subcomponents in a part-to-whole manner. This learning bias gradually accumulates and distills the features of the subcomponents into local parts and then hierarchically into the whole target,

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allowing us to mine the topological relational features of the target in a discriminative multilevel manner. Graph models that use a flat pattern learning method are not suitable for our inductive learning preferences [52]. Second, although ranking-based node selection is an effective approach to identify strong scattering points, it is susceptible to interference, causing some noise to be treated as nodes while ignoring some relatively weak subcomponent scattering. In addition, these distributed scattering centers are sensitive to variations in azimuth angle, resulting in these scattering points being selected as nodes at some azimuth angles and ignored at others. These problems can lead to false and missing nodes in the graphs of the target at each observation angle, which increases the difficulty of graph learning. Consequently, the selected graph learning method should have the ability to generalize and align from unseen nodes, but unfortunately, the above models are incapable to effectively handle the variations and uncertainties present in different target scattering points.

In order to address the aforementioned issues and fulfill the necessary requirements, we have opted for the graph sample and aggregate model (SAGEConv) [53] as our convolution operator for local feature aggregation. Furthermore, we have incorporated the differentiable graph pooling operator (DiffPool) [52] to enable hierarchical part-to-whole learning. By adopting and stacking these two modules, our proposed framework is capable of aggregating feature information from a node's local neighborhood, thereby enabling the learning of local part features. This approach effectively captures the topological relationships between the subcomponents and the target, facilitating the extraction of informative features from the scattering topology domain. Accordingly, we aim to establish a multilayer graph learning framework with the hierarchical structure $\{\mathcal{G}^l \subset \cdots \subset \mathcal{G}^0\}$.[1] In each layer, we learn a coarse graph whose nodes and edges are derived from those in the upper layer based on the topology, which promotes part-to-whole learning bias. More specifically, we represent the feature vectors of the ith node of $\mathcal{G}^l$ in layer l as $\mathbf{h}_i^l \in \mathbb{R}^{d_l}$. SAGEConv is a spatial-based graph convolution operator that computes $\mathbf{h}_i^l$ using the following equation:

$$\mathbf{h}_i^l \leftarrow \frac{\sigma\left(\mathbf{W}_1^l \mathbf{h}_i^{l-1} + \mathbf{W}_2^l \mathbf{h}_{\mathrm{Nei}(i)}^l\right)}{\left\|\sigma\left(\mathbf{W}_1^l \mathbf{h}_i^{l-1} + \mathbf{W}_2^l \mathbf{h}_{\mathrm{Nei}(i)}^l\right)\right\|_2}$$
$$\text{s.t. } \mathbf{h}_{\mathrm{Nei}(i)}^l = \mathrm{AGGREGATE}\left(\mathbf{h}_j^{l-1} \ \forall j \in \mathrm{Nei}(i)\right) \tag{9}$$

where $\mathrm{Nei}(i)$ denotes the neighbor of node i, $\mathbf{W}_1^l$ and $\mathbf{W}_2^l$ are two parameterized tranformation matrices in lth layer. The elementwise mean aggregator is used for aggregation operator, which is the same as in (7). After extracting feature vectors for all nodes in a given graph layer, the DiffPool operator is leveraged to compress the subcomponent information and propagate it to the nodes in the subsequent layer. This operator can learn a cluster assignment matrix $\mathbf{S}^l \in \mathbb{R}^{N_l \times N_{l+1}}$ that maps the nodes' feature matrix $\mathbf{H}^l = [\mathbf{h}_1^l, \ldots, \mathbf{h}_{N_l}^l]^{\mathrm{T}}$ of the current layer to the next layer based on their feature vectors according to the following equation:

$$\mathbf{H}^{l+1} = \left(\mathbf{S}^l\right)^{\mathrm{T}} \mathbf{H}^l. \tag{10}$$

Each element $\mathbf{S}_{i,j}^l$ represents the probability that the ith node in the lth layer belongs to the jth node in the next layer. The resulting coarsened graph retains the essential information from the previous subcomponent graph. The corresponding coarsened adjacency matrix $\mathbf{A}^{l+1}$ in the next layer is also computed via the following equation:

$$\mathbf{A}^{l+1} = \left(\mathbf{S}^l\right)^{\mathrm{T}} \mathbf{A}^l \mathbf{S}^l. \tag{11}$$

The coarsened adjacency matrix $\mathbf{A}^{l+1}$ and the cluster embeddings $\mathbf{H}^{l+1}$ are utilized as inputs to a subsequent SAGEConv layer, which generates high-level embeddings for each node. The DiffPool operator is employed to optimize the node clustering process. This operator involves two inner optimization objectives. The first objective is the auxiliary link prediction objective, which aims to cluster adjacent nodes into the same cluster as much as possible. The second objective is to minimize disorder by avoiding each cluster containing the same node through entropy minimization. These two optimization objectives are formulated in the following equation:

$$\mathcal{L}_{\mathcal{G}} = \left\|\mathbf{A}^l - \mathbf{S}^l\left(\mathbf{S}^l\right)^{\mathrm{T}}\right\|_{\mathrm{F}}^2 + \frac{1}{n}\sum_{i=1}^{N_l} \mathbf{H}\left(\mathbf{S}_i^l\right) \tag{12}$$

where $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm, $\mathbf{H}$ denotes the entropy function, and $\mathbf{S}_i^l$ denotes the ith row of the cluster assignment matrix $\mathbf{S}^l$. Multiple SAGEConv and DiffPool are stacked layer-by-layer to generate final discriminative features $\mathbf{f}_s \in \mathbb{R}^d$ that represent the discriminative topological relationships among the strong scatter points.

C. Image-Domain Feature Extraction Module

In Sections III-A and III-B, we presented two modules for discriminative feature learning from the phase-history domain and scattering point domain, respectively. In this section, we focus on extracting image features of the SAR target. However, developing a new learning architecture for SAR target image is not within the scope of this article. Instead, we propose that any off-the-shelf image feature extractor, such as VGG [44] or ResNet [54], can be utilized as the backbone. For the sake of simplicity, we choose the feature extractor of the widely used baseline model A-ConvNet [16] as the backbone which is specifically designed for SAR ATR. We use the first five layers of A-ConvNet as the image feature extractor.

D. Multimodal Feature Discriminative Fusion Module

We have developed three feature learning modules for three different domains, namely, phase history, scattering points, and images. To integrate the three feature vectors extracted from these domains, we employ a concatenation-based fusion method. Specifically, each feature extraction module maps the input from different domains to the corresponding feature space using three extractors $\mathbf{g}_p$, $\mathbf{g}_s$, and $\mathbf{g}_i$, respectively, as

$$\mathbf{f}_p = \mathbf{g}_p(\mathbf{r}), \quad \mathbf{f}_s = \mathbf{g}_s\left(\mathcal{G}^0\right), \ \mathbf{f}_I = \mathbf{g}_i(\mathbf{X}) \tag{13}$$

[1]The notation here does not mean subgraph relation, but hierarchical structure via node pooling operation.

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TABLE I
DETAILED ARCHITECTURE INFORMATION OF EACH MODULE

Module	Architecture	Input	Output
$\mathbf{g}_p$	Conv*2+MaxPool:K1-S1-C4-K3-S1-C8-ReLu+P2-S2 Conv+MaxPool:K3-S1-C16-ReLu+P2-S2 Conv+MaxPool:K3-S1-C32-ReLu+P2-S2 Conv+MaxPool:K3-S1-C64-ReLu+P2-S2 Conv: K3-S1-C128 TransformerEncoderLayer:C128-nhead2 LayerNorm	$\mathbf{r}$	$\mathbf{f}_p$
$\mathbf{g}_s$	SAGEConv*2+DiffPool:C16-ReLu-C32-ReLu+C32-N30 SAGEConv*2+DiffPool:C32-ReLu-C64-ReLu+C64-N10 SAGEConv+DiffPool:C64-ReLu+C64-N1 SAGEConv:C128 LayerNorm	$\mathcal{G}^0$	$\mathbf{f}_s$
$\mathbf{g}_i$	Conv+MaxPool:K5-S1-C16-ReLu+P2-S2 Conv+MaxPool:K5-S1-C32-ReLu+P2-S2 Conv+MaxPool:K6-S1-C64-ReLu+P2-S2 Dropout Conv:K5-S1-C128-ReLu Conv:K3-S1-C128 LayerNorm	$\mathbf{X}$	$\mathbf{f}_i$
$\mathbf{g}_f$	Conv:K1-S1-C128-ReLu Conv:K1-S1-CC	$[\mathbf{f}_p, \mathbf{f}_s, \mathbf{f}_I]$	$\mathbf{f}$

where $\mathbf{r}$, $\mathcal{G}^0$, and $\mathbf{X}$ are phase-history raw feature vector, scattering points graph, and the magnitude image of an SAR target, respectively. The resulting feature vectors, $\mathbf{f}_p$, $\mathbf{f}_s$, and $\mathbf{f}_I$, are normalized and concatenated to mitigate the impact of distribution discrepancy. The concatenated feature vector is then fed into a feature fusion operator $\mathbf{g}_f$ to obtain the fused features for final target recognition. This fusion strategy combines complementary information from different modal domains, resulting in a more comprehensive representation of the target.

To further enhance the discriminative power of the feature vectors of each modality and prevent them from collapsing, we introduce a modalwise discriminative regularization module. It aims to ensure that each modular feature vector can also independently and accurately recognize the target by feeding them into three label constrained classifiers that minimize their respective classification error. The overall optimization objective function of the proposed MoFFL can be summarized as follows:

$$\min_{\mathbf{g}_p,\mathbf{g}_s,\mathbf{g}_s,\mathbf{g}_f} \sum_{i=1}^{m} \lambda\big(\ell\big(\mathbf{g}_p(\mathbf{r}_i),\mathbf{y}_i\big)+\ell\big(\mathbf{g}_s\big(\mathcal{G}_i^0\big),\mathbf{y}_i\big)+\ell(\mathbf{g}_i(\mathbf{X}_i),\mathbf{y}_i)\big) + \ell\big(\mathbf{g}_f\big(\big[\mathbf{f}_p,\mathbf{f}_s,\mathbf{f}_I\big]_i\big),\mathbf{y}_i\big) + \sum_{j=1}^{k}\mathcal{L}_{\mathcal{G}} \tag{14}$$

where $\mathbf{y}$ refers to the label assigned to the ith sample during training, ℓ is the cross-entropy loss function, the subscript notation denotes the specific training sample, k represents the number of layers in the DiffPool operator used in the scattering topological domain feature extraction module, and λ is a hyperparameter that controls the strength of the modalwise discriminative regularization module. For detailed information about the architecture of each module in MoFFL, please refer to Table I.

IV. EXPERIMENTS AND ANALYSIS

This section presents extensive experiments conducted to demonstrate the effectiveness and superiority of our proposed framework for SAR ATR. First, we introduce the MSTAR dataset used for the experiments along with the experimental settings. Second, we perform ablation experiments to validate the effectiveness of our proposed framework. Finally, we compare the proposed MoFFL with other SAR ATR models to demonstrate its superiority.

TABLE II
INFORMATION OF RADAR PARAMETERS OF MSTAR DATABASE

Range 3dB Width=0.302400
Cross Range 3dB Width=0.321600
Center Frequency=9.60GHz
Cross Range Weighting=-35dB_Taylor
Range Weighting=-35dB_Taylor
Dynamic Range=64dB
Bandwidth=0.591GHz
Range Resolution=0.304700
Cross Range Resolution=0.304700
Range Pixel Spacing=0.202148
Cross Range Pixel Spacing=0.203125

A. Experimental Settings

1) Dataset Settings: The MSTAR dataset, obtained using Sandia National Laboratory's X-band SAR sensor platform, is a widely recognized benchmark for evaluating the performance of SAR ATR systems. It comprises ten categories of publicly released military ground targets, each with varying azimuth and depression angles, and an image resolution of 0.3 × 0.3 m. The targets include armored transport vehicles such as BMP-2, BRDM-2, BTR-60, and BTR-70, a rocket launcher (2S1), tanks (T-62, T-72), an air defense unit (ZSU234), a truck (ZIL-131), and a bulldozer (D7). Fig. 5 displays optical images of the ten target types for illustrative purposes. Following common practice [16], centrally cropped image patches of 88×88 magnitude are employed for training and testing. The initial feature $\mathbf{r}$ in the phase-history domain is computed using parameters reflecting radar characteristics, as summarized in Table II based on the MSTAR header file. To demonstrate the superiority of our proposed framework with limited training samples, we only randomly selected a portion (up to 40% for each class) of target images from the training set for each sample to be used for model learning. It is worth noting that, unlike prior deep learning-based SAR ATR approaches, no other data augmentation techniques were utilized during model training. While data augmentation is certainly an effective strategy to improve generalization and provide further gains, our primary motivation of this study is to explore how far we can improve the performance through exploiting the intrinsic multimodal information from the real original SAR data without external manipulations.

In order to comprehensively assess the performance of this model, different testing conditions, called the standard operating conditions (SOCs) and several extended operating conditions (EOCs), will be tested. Specifically, the SOC scenario refers to that the identical serial numbers and configurations in the test set are the same as those in the training set, but with different aspects and depression angels. Images for training are captured at 17° but at 15° for testing. In EOC scenario, there are large differences including substantial variations in depression angle, target configuration, and version variants between the train set and the test set. The detailed

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Fig. 5. Optical sample images of ten types of target in MSTAR database only for illustration: (a) BMP2; (b) BTR70; (c) T72; (d) 2S1; (e) BRDM2; (f) BTR60; (g) D7; (h) T62; (i) ZIL131; and (j) ZSU234.

TABLE III
INFORMATION OF SOC

		Training Set		Testing Set	
Class	Serial No.	Depression	No.	Depression	No.
BMP-2	9563,9566,C21	17°	698	15°	587
BTR-70	c71	17°	233	15°	196
T-72	132,812,S7	17°	691	15°	582
BTR-60	k10yt7532	17°	256	15°	195
2S1	b01	17°	299	15°	274
BRDM-2	E-71	17°	298	15°	274
D7	92v13105	17°	299	15°	274
T-62	A51	17°	299	15°	273
ZIL-131	E12	17°	299	15°	274
ZSU-234	d08	17°	299	15°	274

TABLE IV
INFORMATION OF EOC-1 (LARGE DEPRESSION VARIATION)

		Training Set		Testing Set	
Class	Serial No.	Depression	No.	Depression	No.
T-72	SN132,A64	17°SN132	232	30°A64	288
2S1	b01	17°	299	30°	288
BRDM-2	E-71	17°	298	30°	287
ZSU-234	d08	17°	299	30°	288

TABLE V
INFORMATION OF EOC-2 (CONFIGURATION VARIATION)

	Class	Serial No.	Depression	No.
Training Set	T72	132	17°	232
	BMP-2	9563	17°	233
	BRDM-2	E-71	17°	298
	BTR-70	c71	17°	233
Testing Set	T72	S7	17°,15°	419
		A32	17°,15°	572
		A62	17°,15°	573
		A63	17°,15°	573
		A64	17°,15°	573

information for SOC and three types EOCs, namely, EOC-1, EOC-2, and EOC-3, is presented in Tables III–VI, respectively.

2) Experimental Settings: Our framework employs default parameter initialization in each layer, a mini-batch size of 32, and utilizes the standard Adam optimizer with a learning rate of 0.0001 and default other hyperparameters. The hyperparameter λ is set to 1. To ensure result reliability, we conduct five experiments and report the average accuracy. All experiments are conducted using PyTorch 1.13 library on an NVIDIA GeForce RTX 3070 GPU.

B. Ablation Experiments

In this section, we perform ablation experiments to validate the effectiveness of MoFFL and investigate the complementarity of the multimodal features. More specifically, MoFFL consists of three modules: the phase-history domain feature extraction module (denoted as Model_p), the scattering topology domain feature extraction module (denoted as Model_s), and the image-domain feature extraction module (denoted as Model_i). Each of them will be evaluated in the following part.

TABLE VI
INFORMATION OF EOC-3 (VERSION VARIATION)

	Class	Serial No.	Depression	No.
Training Set	T72	132	17°	232
	BMP-2	9563	17°	233
	BRDM-2	E-71	17°	298
	BTR-70	c71	17°	233
Testing Set	T72	812	17°,15°	426
		A04	17°,15°	573
		A05	17°,15°	573
		A07	17°,15°	573
		A10	17°,15°	567
	BMP-2	9566	17°,15°	428
		c21	17°,15°	429

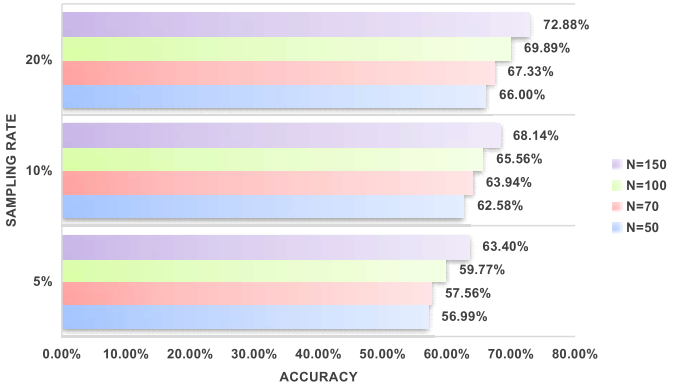


Fig. 6. Comparison results of different number of strong scattering points N on SOC.

First, we conducted experiments on Model_s to examine the effect of the number of extracted strong scattering points on recognition accuracy. Specifically, we varied the number of strong scattering points extracted (denoted as N) to investigate their impact on recognition accuracy. We conducted experiments with training sample rates of 5%, 10%, and 20%, respectively, and extracted strong scattering points with $N = \{50, 70, 100, 150\}$. We then utilized these strong scattering points to construct the initial scattering graph $\mathcal{G}$ and input it into Model_s to evaluate their recognition accuracy results, as presented in Fig. 6. The results of our experiments indicate that recognition accuracy generally improves with an increase in both the number of extracted strong scattering points and the training sampling rate. However, it is crucial to consider the potential drawbacks of extracting too many strong scattering points, such as the inclusion of background clutter and increased computational cost during the training and testing phases. Therefore, to optimize the tradeoff between recognition accuracy and computational cost, we chose $N = 70$ as the number of extracted strong scattering points. Our evaluation demonstrates that this value effectively captures the key points on the target subcomponents while minimizing the impact of background clutter.

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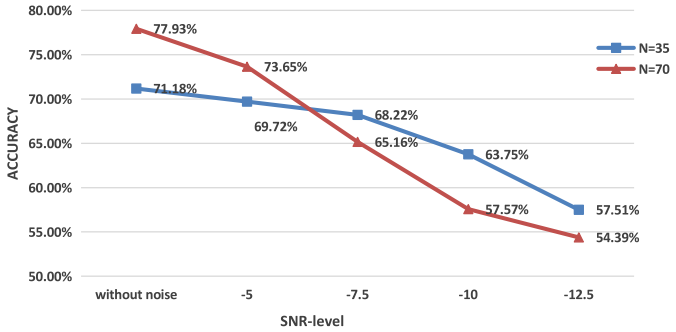


Fig. 7. Comparison results of different number of strong scattering points N across the level of noise.

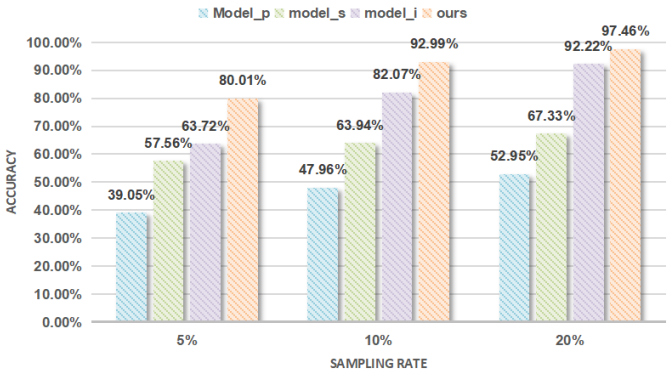


Fig. 8. Comparison of the phase-history domain feature extraction module Model_p, the scattering topology domain feature extraction module Model_s, image-domain feature extraction module Model_i, and the feature fusion model on recognition accuracy on SOC.

Next, we introduced frequency-modulated interference with different signal-to-noise ratios (SNRs) to the phase-history data of the test dataset to investigate the impact of noise on the selection of strong scattering points. Specifically, with SNRs of -5, -7.5, -10, and -12.5 dB, we extracted 35 and 70 strong scattering points to construct the graph $\mathcal{G}$. We then fed this graph into the Model_s, which was trained using the same number of strong scattering points without noise, to obtain the recognition accuracy. The results are depicted in Fig. 7. From the experimental findings, it can be deduced that the recognition accuracy decreases as the SNR decreases. In scenarios where no additional noise is added to the test data or when the noise level is low, extracting more strong scattering points leads to higher recognition accuracy. Conversely, when the noise level is high, the Model_s that extracts fewer strong scattering points performs better. This experiment illustrates the relationship between the number of extracted strong scattering points and the test scenario. In situations where the background clutter is significant, it is crucial to appropriately reduce the number of extracted strong scattering points to ensure their reliability. Furthermore, we evaluated the robustness of the Model_s by testing the baseline model A-ConvNet under the same experimental setup. The results indicated that the recognition accuracy of the baseline model was only 9.9% under an SNR of -5 dB. This demonstrates that our model exhibits greater robustness in the presence of noise interference.

We proceeded to conduct ablation experiments for each of the three modules to confirm their effectiveness. The network structure of these three modules is identical to that of the multimodal feature fusion network. We trained these three networks on the SOC with sampling rates of 5%, 10%, and 20%, respectively, and report the recognition results in Fig. 8. The results in Fig. 8 show that the recognition accuracy of each individual module in MoFFL is lower than the fused recognition accuracy. Specifically, for the phase-history domain feature extraction module, the recognition accuracy decreases by 40.96%, 45.03%, and 44.51% when the sampling rate is 5%, 10%, and 20%, respectively. This decrease in recognition accuracy is mainly due to network overfitting caused by rather insufficient training samples (about ten samples/class for 5% case). For the scattering topology domain feature extraction module, the recognition accuracy decreases by 22.45%, 29.05%, and 30.13%, respectively, mainly due to the loss of target contour information caused by limited strong scattering points. In addition, for the image-domain feature extraction module, the recognition accuracy decreases by 16.29%, 10.92%, and 5.24%, respectively, and the difference between the recognition accuracy and the recognition accuracy after feature fusion decreases with the increase of the sampling rate. Nevertheless, the results clearly show that the fused recognition accuracy is higher than that of the individual modules at all three sampling rates, indicating that the phase-history domain features, the scattering topology domain features, and the image-domain features complement each other. The fusion of these three modal features can lead to more complete discriminative target features. This observation supports our motivation that no single feature modular, such as image or ASC features, is sufficient to represent SAR targets. Instead, the radar-related features implied in the phase-history domain features also reflect target characteristics.

C. Framework Comparison on SAR ATR

To validate the superiority, we compare the MoFFL with other SAR ATR algorithms on different operating conditions.

1) Results of SOC: First, we conduct a comparison experiment on the SOC condition, where the training set and test set information are detailed in Table I. To validate the recognition performance of our proposed framework, we selected several previously proposed classical SAR ATR algorithms as comparison methods with limited training data, where the sampling rate ranges from 5% to 40%. The comparison models include A-ConvNet [16], VGG-Net [44], ResNet50-V2 [54], AlexNet [45], SDF-Net [34], KNN, linear kernel support vector machine (L-SVM), and sparse representation classifier (SRC) [55]. Among these methods, the first four are well-known deep learning algorithms that perform well in SAR ATR tasks, SDF-Net is a scatter center feature and deep feature fusion learning algorithm, and the last three are classical ATR algorithms. The accuracy curves of different algorithms are shown in Fig. 9. Furthermore, we present the confusion matrix of our framework with 40% training samples in Table VII to provide a more detailed analysis of the recognition performance.

Based on the comparison results, our proposed method outperforms all the comparison methods in terms of recognition

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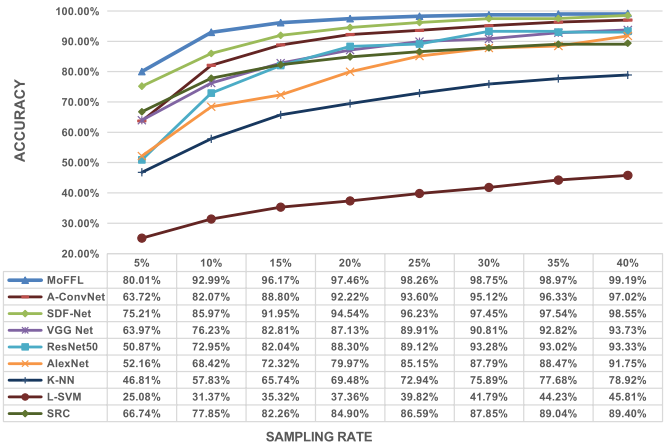


Fig. 9. Comparison results of different SAR ATR frameworks on SOC.

TABLE VII

CONFUSION MATRIX OF FIVE TIMES RESULTS WITH 40% TRAINING SAMPLES

	2S1	BMP-2	BRDM-2	BTR-60	BTR-70	D7	T-62	T-72	ZIL-131	ZSU-234
2S1	1347	0	10	0	0	0	8	0	0	5
BMP-2	0	2918	5	3	1	0	0	8	0	0
BRDM-2	0	15	1347	0	1	0	0	7	0	0
BTR-60	0	0	3	956	14	0	0	2	0	0
BTR-70	1	4	1	1	973	0	0	0	0	0
D7	0	0	0	0	0	1360	0	0	10	0
T-62	6	0	0	0	0	5	1353	0	0	1
T-72	0	11	0	2	0	0	0	2897	0	0
ZIL-131	0	0	0	0	0	2	0	0	1368	0
ZSU-234	0	3	0	0	0	0	0	0	0	1367

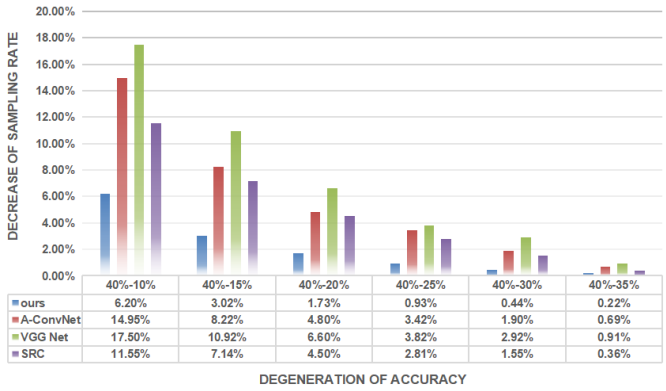


Fig. 10. Accuracy degeneration comparison as sampling rate decreases.

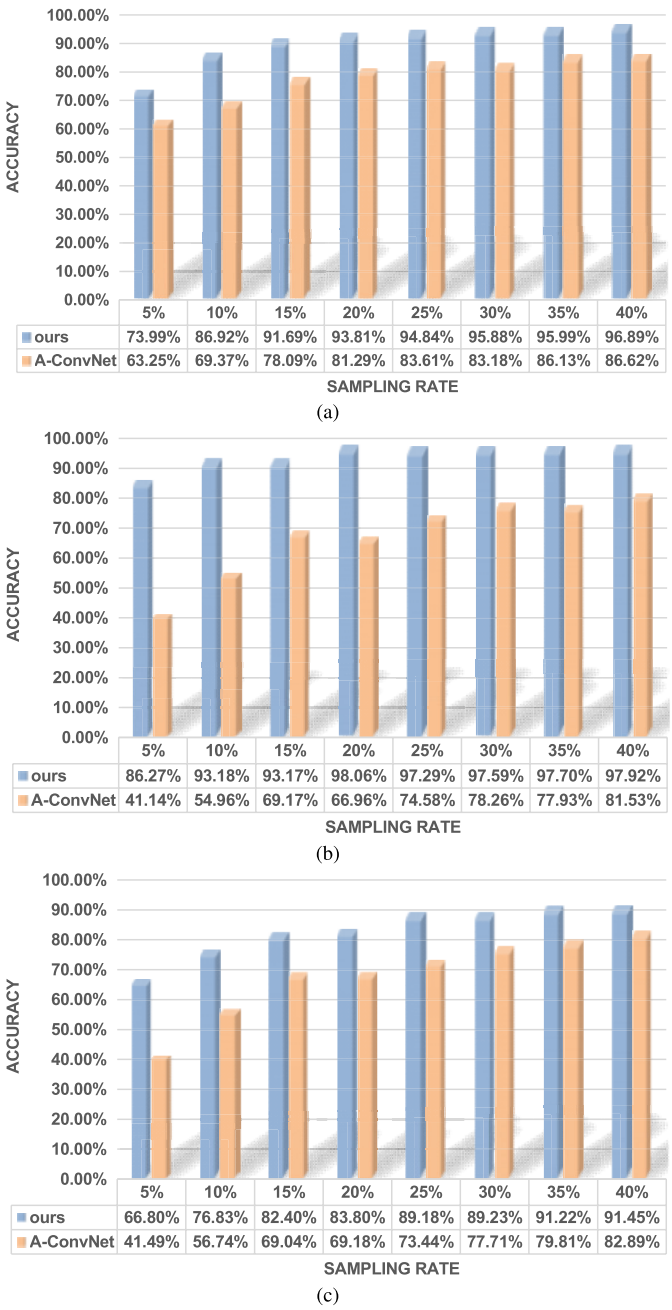


Fig. 11. Comparison results on different EOCs: (a) EOC-1; (b) EOC-2; and (c) EOC-3.

accuracy at all sampling rates. In particular, at a sampling rate of 5%, our method achieves a recognition accuracy of 80.01% that is 16.29% higher than that of A-ConvNet, indicating that the incorporation of phase-history domain features and scattering topology domain features provides additional target supervised information. While the recognition accuracy of SDF-Net is also commendable, our framework achieves slightly better performance due to the additional fusion of phase-history domain features. In contrast, VGG-Net, ResNet50, and AlexNet show poor recognition results because they do not take into account the imaging differences between optical and SAR images. SRC, which does not require a learning process, outperforms A-ConvNet in the case of insufficient samples. Conversely, K-NN and L-SVM show the lowest accuracy in this task.

To evaluate the robustness of our proposed framework to reduced training samples, we investigate the degradation of recognition accuracy from a sampling rate of 40%–10%. Fig. 10 shows the results of this investigation. Fig. 10 shows that our proposed framework exhibits the highest robustness in all cases with reduced training samples, due to its ability to extract more comprehensive target features. The next most robust model is the SRC model, which can be considered as a parameter-free model. Conversely, A-ConvNet and VGG-Net rely solely on SAR image information to extract discriminative features, which are prone to overfitting when training samples are insufficient, resulting in poor robustness.

2) Results of EOC: Finally, we designed experiments to test our proposed framework under three EOC conditions: EOC-1, EOC-2, and EOC-3, as shown in Tables IV–VI,

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respectively, to verify its superiority under large depression variation, configuration variation, and version variation compared to the A-ConvNet benchmark. The comparison results are shown in Fig. 11. It is clear from the three figures that our framework outperforms A-ConvNet in terms of recognition accuracy at different sampling rates. Moreover, our framework shows robust superiority after fusing the phase-history domain features and the scattering topology domain features on the A-ConvNet benchmark. In particular, under the EOC-2 framework, the recognition accuracy of our framework is more than 45% higher than that of A-ConvNet when the sampling rate is 5%. This observation highlights the superiority of our framework in the case of configuration variation. Overall, our proposed framework exhibits superior performance in the case of large depression variation, configuration variation, and version variation.

V. Conclusion

In conclusion, this article proposes a novel MoFFL for SAR ATR, which aims to capture the full range of radar characteristics, scattering topology, and image features from different SAR target data domains and achieve feature fusion for enhanced complementarity. The proposed framework uses an encoding scheme and a cross-directional sequence learning module to extract discriminative features from the phase-history domain, a hierarchical graph node aggregating neural network to capture scattering topology features, and a standard feature extractor for image features. The experimental results on the MSTAR benchmark demonstrate the effectiveness of our proposed approach, especially for learning with limited samples. Our model achieves the state-of-the-art recognition accuracy for the SAR ATR task, outperforming other ATR algorithms. The proposed framework provides a promising direction for improving the performance of SAR ATR systems.

It is important to highlight that the proposed framework currently assumes a single target in the illuminated scene and encodes radar characteristics into the raw feature vector of the phase-history domain. Extending the framework to handle multiple targets in the phase-history domain is an area we plan to explore in our future work. On another note, SAR ATR is progressing toward interferometry SAR/inverse synthetic aperture radar (ISAR) and polarimetric interferometry SAR/ISAR (InISAR). In this context, the 3-D polarimetric InISAR imaging of noncooperative targets [56] holds the potential to generate a 3-D point cloud of the target, which introduces new modal information to our research. This aspect will be a focus of our future investigations.

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